

LAB 3 - COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Problem statement: Blood Bank Emergency Matcher

Name: Muruli H

SRN: PES1UG24CS676

Scenario Review

The Blood Bank Emergency Matcher is designed for emergency situations where blood is required quickly. An Emergency Requester can submit a blood request by specifying the blood group and required units, check whether the required blood is available, and receive the request status. A Blood Bank Manager maintains the blood inventory and issues available blood units for approved requests. If the required blood is unavailable, the system finds suitable donors through emergency donor matching.

1. Requirements Considered

ID	Type	Requirement
FR-01	Functional	Submit an emergency blood request with blood group and required units.
FR-02	Functional	Check availability of the required blood group and units.
FR-03	Functional	Receive the status of the blood request: accepted, rejected, or being processed.
FR-04	Functional	Allow the Blood Bank Manager to maintain blood stock and issue available units for approved requests.
FR-05	Functional	Find suitable blood donors when the required blood is unavailable in the blood bank.
NFR-01	Security	Restrict protected request, inventory and donor information to authorized users.
NFR-02	Performance	Emergency request, availability and matching operations should respond in a timely manner.

2. Architectural Style Analysis

Layered Architecture

Advantages: Clear separation of presentation, business logic and data; simple to understand and maintain.

Limitations: Communication overhead between layers; less independent scaling.

Decision: Selected.

Microservices

Advantages: Independent deployment, scaling and fault isolation.

Limitations: Operational complexity, network latency and data-consistency challenges.

Decision: Not selected.

Client-Server

Advantages: Centralized control and straightforward data consistency.

Limitations: Central server can become a bottleneck and single point of failure.

Decision: Not selected.

3. Architecture Selection

Selected Architecture: Layered Architecture. We chose Layered Architecture because the Blood Bank Emergency Matcher naturally separates user-facing functions, business processing and data storage. This structure is simple to understand and maintain and is suitable for the current system scope.

Two scenario-related reasons: (1) requester, manager and donor interactions can remain in the presentation layer while request, inventory and matching logic stay in the business layer; (2) blood requests, inventory and donor information can be managed in a separate data layer.

Security advantage: the business layer can validate and authorize access before protected blood-bank or donor information is read or changed.

Performance benefit: request processing, availability checking, inventory operations and donor matching are separated into focused components, reducing unnecessary work in the user interface.

4. Components

The components below correspond to the supplied UML component diagram and describe the responsibility of each module.

NFR-02	Performance
User Interface (Web / Mobile App)	Allows users to submit emergency requests, check blood availability, view request status, search donors and manage their profile.
Blood Request Management	Creates and validates requests, updates request status and provides request details.
Blood Inventory Management	Checks blood availability, updates inventory, reserves/releases units and provides inventory details.
Emergency Donor Matching	Finds matching donors, retrieves donor details, notifies donors and manages donor responses.
Blood Bank Database	Stores and retrieves blood requests, inventory information and donor information.

How the components fit together: The user-facing interface connects to Blood Request Management. Blood Request Management communicates with Blood Inventory Management to check availability and can invoke Emergency Donor Matching when blood is unavailable. The inventory and donor components use the Blood Bank Database for the information required by their operations.

5. Component Diagram

The following is the component diagram supplied for the Blood Bank Emergency Matcher. It is used here without changing the diagram, so the report remains consistent with the project model.

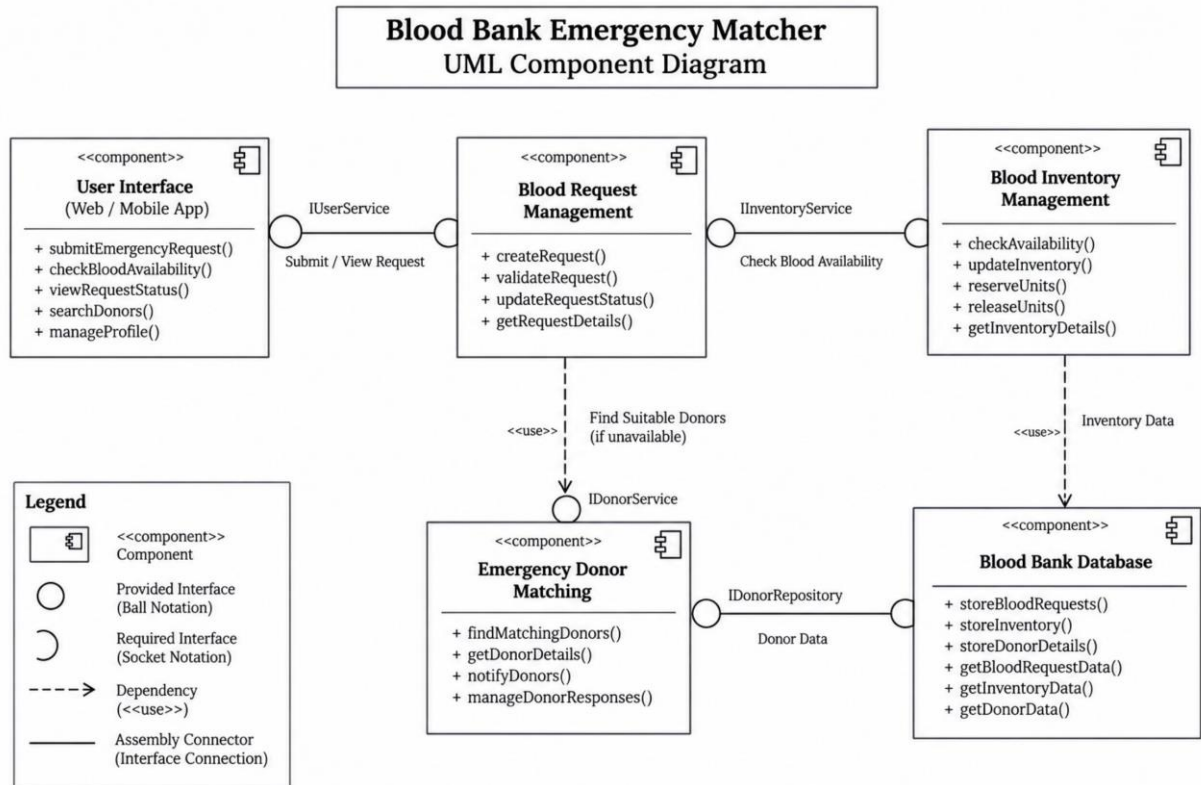


Diagram notation: component rectangles represent modular components. The ball-and-socket symbols show provided and required interfaces, while the labelled connectors show communication or dependency between components.

6. Interfaces and Interactions

The following interfaces and connections are taken from the supplied component diagram.

Blood Bank Database	Stores and retrieves blood requests, inventory information and donor information.	Emergency request, availability and matching operations should respond in a timely manner.
IUserService	User Interface → Blood Request Management	Submit and view emergency requests.
IInventoryService	Blood Request Management ↔ Blood Inventory Management	Check blood availability.
IDonorService	Blood Request Management → Emergency Donor Matching	Find suitable donors when blood is unavailable.
IDonorRepository	Emergency Donor Matching ↔ Blood Bank Database	Access donor data.
Inventory Data	Blood Inventory Management → Blood Bank Database	Read and update inventory data.
Request Data	Blood Request Management → Blood Bank Database	Store and retrieve request data.

7. Security and Performance

Security: Blood requests, inventory records and donor information should be accessible only to authorized users. The layered design keeps these records behind application/business components, where authorization and validation can be applied before data is returned or modified.

Performance: Emergency request, availability and matching operations should be processed with minimal delay. Separating these responsibilities into focused components helps avoid unnecessary processing and keeps each operation close to the component responsible for it.

8. Report Summary

The Blood Bank Emergency Matcher is modelled using Layered Architecture. The component model separates the user interface from request management, inventory management and emergency donor matching, with the Blood Bank Database supporting data storage. The interfaces show how these components communicate to support emergency blood requests, availability checking, request status, inventory maintenance, blood issuing and donor matching. The design also addresses security and performance considerations.